

Herbicide Treatment Table

	Common name	Amount per acre	REI‡	PHI‡
	(Example trade name)		(hours)	(days)
Pesticide precautions Protect water Calculate VOCs Protect bees				
Not all registered pesticides are listed. The following are listed alphabetically. When choosing a pesticide, consider information relating to environmental impact(/mitigation/), resistance management(/IPMPROJECT/glyphosateresistance.html), the pesticide's properties, and application timing. Tank mixes may be necessary to achieve desired control; see the Susceptibility of Weeds to Herbicide Control(/agriculture/corn/susceptibility-of-weeds-to-herbicide-control/) for information on specific weed control. Always read the label of the product being used.				
BEFORE PLANTING				
	ATRAZINE	1–2 lb a.i.	12	60
	(GCS Atrazine 4L)	2–4 pt		
WSSA MODE OF ACTION GROUP NUMBER ¹ : 5				
COMMENTS: Band application can reduce carryover. Considered to be a ground water contaminant and requires a use permit within Ground Water Protection Areas. No fall applications are permitted,				
	EPTC	6 lb a.i.		
	(Eptam 20-G Granules)	30 lb	12	NA
WSSA MODE OF ACTION GROUP NUMBER ¹ : 8				
COMMENTS: Immediate incorporation is necessary. Use high rates for perennial weeds. Requires cross-discing for best results. Some corn varieties are sensitive to injury with this material; check with corn seed companies to ensure a variety is tolerant before using. Soils lacking enough moisture for seed germination must be pre-irrigated before application.				
	GLYPHOSATE	4.54 lb a.i.		
	(Roundup WeatherMAX)	3.3 qt	4	NA
WSSA MODE OF ACTION GROUP NUMBER ¹ : 9				
COMMENTS: A nonselective herbicide; apply to emerged weeds. Use with a nonionic surfactant.				

	Common name	Amount per acre	REI‡	PHI‡
	(Example trade name)		(hours)	(days)
	PARAQUAT*	0.57–1.39 lb a.i.		
	(Gramoxone SL 3.0)	1.3–2.7 pt (preplant)	48	NA
	WSSA MODE OF ACTION GROUP NUMBER ¹ : 22			
	COMMENTS: Apply as a contact broadcast spray to emerged weeds. Use with a nonionic surfactant.			
	S-METOLACHLOR	0.837–2.48 lb a.i.		
	(Dual Magnum)	1–2.6 pt	24	NA
	WSSA MODE OF ACTION GROUP NUMBER ¹ : 15			
	COMMENTS: Apply after preirrigation and incorporate into the soil.			
	AT PLANTING			
	ATRAZINE	1.6–2 lb a.i.		
	(GCS Atrazine 4L)	3.2–4 pt	12	See comments
	WSSA MODE OF ACTION GROUP NUMBER ¹ : 5			
	COMMENTS: Band application can reduce carryover. PHI is 45 days for sweet corn and 60 days for field corn. Considered to be a ground water contaminant and requires a use permit within Ground Water Protection Areas.			
	GLYPHOSATE	1.375–4.5375 lb a.i.		
	(Roundup WeatherMAX)	1–3.3 qt	4	NA
	WSSA MODE OF ACTION GROUP NUMBER ¹ : 9			
	COMMENTS: Apply to emerged weeds but before corn emerges. Use with a nonionic surfactant.			
	PARAQUAT*	0.57–1.39 lb a.i.		
	(Gramoxone SL 3.0)	1.3–2.7 pt	48	NA
	WSSA MODE OF ACTION GROUP NUMBER ¹ : 22			
	COMMENTS: Apply as a contact broadcast spray to emerged weeds. Use with a nonionic surfactant.			
	PENDIMETHALIN	Label rates		
	(Prowl H2O)	2–4 pt	24	NA
	WSSA MODE OF ACTION GROUP NUMBER ¹ : 3			
	COMMENTS: Irrigate to activate; do not incorporate. Do not apply to no-till corn. Plant at least 1.5 inches deep. Do not use on peat soils.			

	Common name	Amount per acre	REI‡	PHI‡
	(Example trade name)		(hours)	(days)
	S-METOLACHLOR	1.42875–2.48 lb a.i.		
	(Dual Magnum)	1.5–2.6 pt	24	NA
	WSSA MODE OF ACTION GROUP NUMBER ¹ : 15			
	COMMENTS: Apply after preirrigation and incorporate into soil.			
	AFTER PLANTING			
	2,4-D*	0.2375–0.475 lb a.i.		
	(Amine)	0.5–1 pt	48	See comments
	WSSA MODE OF ACTION GROUP NUMBER ¹ : 4			
	COMMENTS: Apply from crop emergence to tasseling. Apply over the top when corn is less than 8 inches tall. Use drop nozzles when corn is more than 8 inches tall, keeping spray mist out of whorl. Don't use when corn is over 3 feet tall or when brace roots have formed. Delay cultivation 8-10 days to prevent stalk breakage due to temporary brittleness caused by 2,4-D. PHI for field and popcorn is 7 days; for sweet corn it is 45 days.			
	ATRAZINE	1.2–2 lb a.i.		
	(GCS Atrazine 4L)	2.4–4 pt	12	See comments
	WSSA MODE OF ACTION GROUP NUMBER ¹ : 5			
	COMMENTS: Do not rotate to any crop except corn or sorghum or injury will occur. Band application can reduce carryover. PHI for Atrazine is 45 days for sweet corn and 60 days for field corn. Considered to be a ground water contaminant and requires a use permit within Ground Water Protection Areas.			
	BROMOXYNIL	0.128–0.55 lb a.i.		
	(Maestro 4EC)	0.5–1 pt	24	45
	WSSA MODE OF ACTION GROUP NUMBER ¹ : 6			
	COMMENTS: For field corn and popcorn only. Apply after crop emergence but before tassels emerge. Weed should be less than 4 inches tall. Can be used where drift of dicamba or 2,4-D may injure sensitive broadleaf crops. Combinations of bromoxynil with other broadleaf herbicides enhance weed control.			
	CARFENTRAZONE	0.03 lb a.i.		
	(Shark EW)	2 fl oz	12	See label
	WSSA MODE OF ACTION GROUP NUMBER ¹ : E			
	COMMENTS: Apply in all tillage systems up to 30 days before planting and afterwards up to 8-leaf stage of corn. For best performance apply to weeds that are 4 inches or shorter. However, can be applied to velvetleaf up to 18 inches tall, or 36 inches tall with drop nozzles. Do not apply to stressed crops or use with an oil concentrate or severe speckled leaf burn may occur. Tank mixing with other herbicides will widen the weed control spectrum.			

	Common name	Amount per acre	REI‡	PHI‡
	(Example trade name)		(hours)	(days)
	DICAMBA*	0.5 lb a.e.		
	(Clarity)	16 fl oz	24	See label
WSSA MODE OF ACTION GROUP NUMBER ¹ : 4				
COMMENTS: Do not use on sweet corn. Apply over the top when corn is less than 8 inches tall. Use drop nozzles when corn is more than 5 leaves or 8 inches tall. Apply postemergence until corn is 36 inches tall or until 15 days before tassels emerge. Keep herbicide out of the whorl of the plant to avoid yield loss. Do not cultivate within 7 days after application. Applications during rapid growth may result in temporary leaning.				
	GLYPHOSATE			
	(Roundup WeatherMAX)	Label rates	4	See label
WSSA MODE OF ACTION GROUP NUMBER ¹ : 9				
COMMENTS: Can be applied as a spot treatment, but apply before corn silks in conventional corn varieties. Corn in treated area will be killed unless it is a Roundup-tolerant variety. Do not allow spray mist to contact desirable plants. Use with a nonionic surfactant. In Roundup Ready varieties glyphosate can be applied over the top to corn up to the V8 stage of corn or 24 inches. Use drop nozzles if corn is taller than 24 inches. Keep spray out of whorls after corn is 30 inches tall. Rates depend on formulation and weed type and size. Application of 1-2 percent dry ammonium sulfate by weight per 100 gal. of water may increase performance. Use extreme care to prevent drift onto non-Roundup Ready crops.				
	HALOSULFURON	0.031–0.062 lb a.i.		
	(Sandea)	0.66–1.33 oz	24	30
WSSA MODE OF ACTION GROUP NUMBER ¹ : 2				
COMMENTS: For use on field corn or field corn grown for seed. Can be applied to sweet corn or popcorn however at lower use rates. Apply over corn beginning at the spike stage until row closure, or until the height of the corn prevents coverage of weed with spray. To avoid the development of resistance to this herbicide by weeds, do not use this material in consecutive years. Use with a nonionic surfactant.				
	HALOSULFURON/DICAMBA*	0.169–0.338 lb a.i.		
	(Yukon)	4–8 oz	12	See label
WSSA MODE OF ACTION GROUP NUMBER ¹ : 2/4				
COMMENTS: For use on field corn and field corn grown for seed. Can be applied over-the-top or with drop nozzles from the spike through 36 inches.				
	METRIBUZIN	0.094–0.141 lb a.i.		
	(Metribuzin 75 WDG)	2–3 fl oz	12	60
WSSA MODE OF ACTION GROUP NUMBER ¹ : 5				

	Common name	Amount per acre	REI‡	PHI‡
	(Example trade name)		(hours)	(days)
COMMENTS: For use on field corn only. Use drop nozzles for rates above 2 oz/acre. Use postemergence as a tank mix with nicosulfuron, atrazine, dicamba, bromoxynil, or 2,4-D to increase number of species controlled, improve control of larger weeds, or reduce herbicide rates (see label).				
	NICOSULFURON	0.015–0.031 lb a.i.		
	(Accent Q)	0.45–0.90 oz	4	See label
WSSA MODE OF ACTION GROUP NUMBER ¹ : 2				
COMMENTS: Do not apply to sweet corn. Apply postemergent until corn is 20 inches tall or 6 leaf stage of corn, whichever comes first, then use drop nozzles up to 36 inches. Keep spray out of whorl. Injury will occur if applied to stressed corn and it won't effectively control stressed weeds. Deep plowing will reduce risk of carryover. Must use with surfactant or COC (crop oil concentrate). Use a minimum of 15 gpa for best performance. Use higher rates as weed size increases. Nicosulfuron can be tank mixed with dicamba or bromoxynil to broaden the weed control spectrum. This material is best used in conjunction with other methods of control to reduce the risk of the development of resistance by weeds.				
	PARAQUAT*	0.25–0.67 lb a.i.		
	(Gramoxone SL 3.0)	0.7–1.3 pt	48	7
WSSA MODE OF ACTION GROUP NUMBER ¹ : 22				
COMMENTS: Apply to emerged weeds. Use shielded spray when corn is under 10 inches. Do not allow spray mist to contact corn plants. Use a nonionic surfactant.				
	PENDIMETHALIN	0.83–1.98 lb a.i.		
	(Prowl 3.3 EC)	1–4.8 pt	24	21
WSSA MODE OF ACTION GROUP NUMBER ¹ : 3				
COMMENTS: Apply as culti-spray after brace roots are formed. May contribute to lodging. Pendimethalin can be active up to 6 months; check the label for plant-back restrictions. Will not control emerged weeds.				

‡	Restricted entry interval (REI) is the number of hours (unless otherwise noted) from treatment until the treated area can be safely entered without personal protective equipment. Preharvest interval (PHI) is the number of days from treatment to harvest. In some cases, the REI exceeds the PHI. The longer of the two intervals is the minimum time that must elapse before harvest.
*	Permit required from county agricultural commissioner for purchase or use.
¹	Group numbers are assigned by the Weed Science Society of America(http://wssa.net) (WSSA) according to different modes of action. Although weeds may exhibit multiple resistance across many groups, mode-of-action numbers are useful in planning mixtures or rotations of herbicides with different modes of action.
NA	Not applicable.



[S.D. Wright\(https://ucanr.edu/About/DirectorySearch/?facultyid=2495\)](https://ucanr.edu/About/DirectorySearch/?facultyid=2495) (emeritus), UC Cooperative Extension Tulare and Kings counties

W.M. Canevari (emeritus), UC Cooperative Extension San Joaquin County

D.J. Munier, UC Cooperative Extension Butte County

Acknowledgement for Contributions to Weeds

M.L. Campbell-Mathews (emeritus), UC Cooperative Extension Stanislaus County

R.N. Vargas (emeritus), UC Cooperative Extension Madera County

Text Updated: 06/25